# TID Direction-of-Arrival Report

19 January 2026 · HamSCI PSWS Network · 10 MHz WWV

section

***Disclaimer:*** *This analysis is experimental. The psws-drf-tid-tools toolkit is under active development and results may not be accurate. Phase speed and direction estimates are derived from cross-correlation of Doppler shift time series and are subject to aliasing, contamination, and geometric limitations of the station array. Results should be treated as indicative only and cross-checked against independent methods before scientific use.*

## EVENT SUMMARY

| Parameter | Value |
|---|---|
| Event date | 19 January 2026 |
| Analysis window | 00:01:02 – 02:03:02 UTC (2.03 hours) |
| TID type | Large-Scale TID (LSTID) |
| Transmitter | WWV, Fort Collins CO (40.68°N, 105.04°W) |
| Frequency | 10.000 MHz |
| Coordinate system | IPP midpoints (great-circle midpoint, station to WWV) |
| Resample cadence | 60 s |
| Bandpass filter | Disabled (raw mean-subtracted signals) |
| Extraction method | Interactive spline (cwt-prophet via tid_spect_click.py) |
| Max xcorr lag | 40 min (auto from baseline / min speed) |
| Software | psws-drf-tid-tools, git commit b9d02cf |

## STATION ARRAY

| Callsign | Location | Lat | Lon | Grid | Run | Notes |
|---|---|---|---|---|---|---|
| AA6BD | Alabama | 35.06°N | 85.13°W | EM75kb | 3-stn + 4-stn | Clean |
| N6RFM | Texas | 32.94°N | 97.21°W | EM12jw | 3-stn + 4-stn | Clean |
| W7LUX | Arizona | 35.10°N | 111.71°W | DM45dc | 3-stn + 4-stn | Clean |
| AC0G_ND | North Dakota | 46.88°N | 96.83°W | EN16ov | 4-stn only | E-region contamination |

## STATION MAP

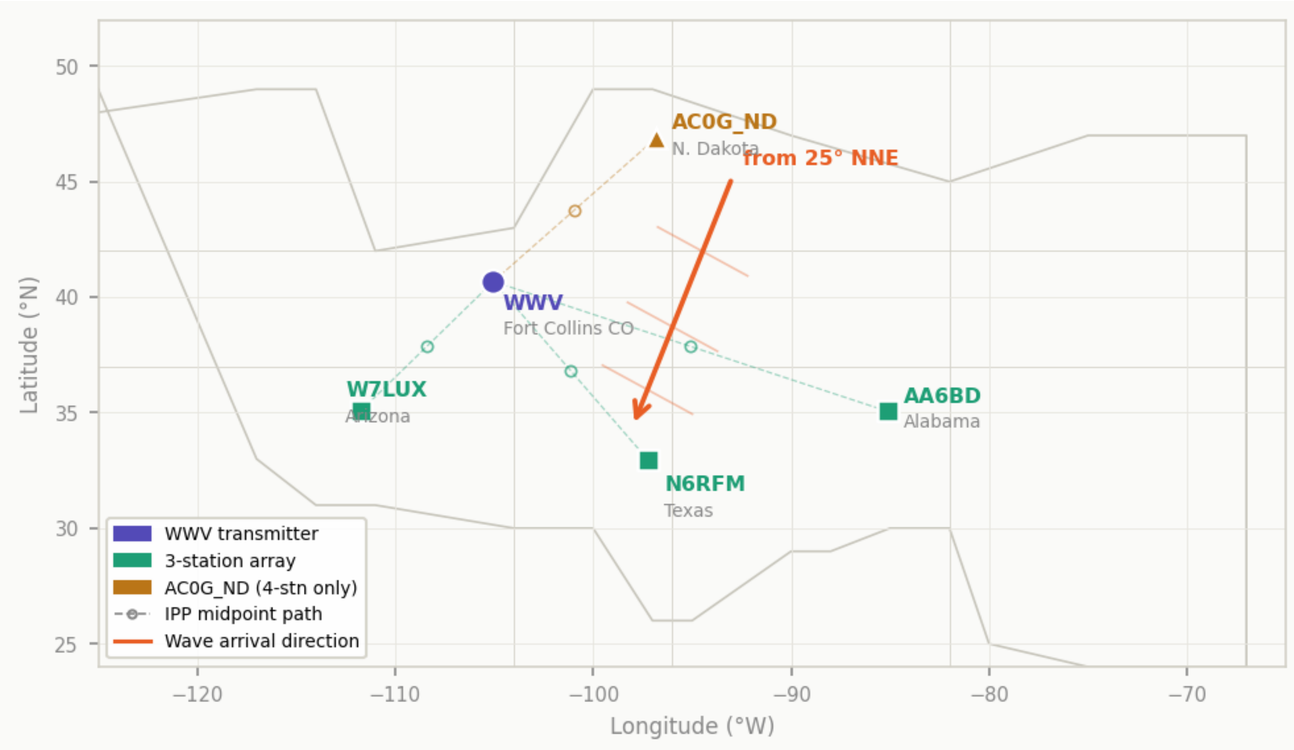


*Figure 1. Station locations relative to WWV transmitter. Dashed lines show WWV–station paths; open circles mark IPP midpoints. Orange arrow indicates wave arrival direction (from 25° NNE). AC0G_ND used in 4-station run only.*

DOA RESULTS

### 3-station run — AA6BD / N6RFM / W7LUX (best result)

| | |
|---|---|
| Phase speed | 341 m/s (1227 km/h) |
| Coming from | 25.3° (NNE) |
| Heading toward | 205.3° (SSW) |
| Wave type | Consistent with large-scale TID (LSTID) |
| Run timestamp | 2026-05-28T173026Z |

### 4-station run — all stations including AC0G_ND (for reference)

| | |
|---|---|
| Phase speed | 445 m/s (1602 km/h) |
| Coming from | 36.2° (NE) |
| Heading toward | 216.2° (SSW) |
| Wave type | Consistent with large-scale TID (LSTID) |
| Run timestamp | 2026-05-28T172939Z |

PAIRWISE CROSS-CORRELATION LAGS

| Pair | Lag (s) | Lag (min) | Correlation | Run | Quality |
|---|---|---|---|---|---|
| AA6BD → N6RFM | +1253 | +20.9 | 0.696 | 3-stn | Good |

| | | | | | |
|---|---|---|---|---|---|
| AA6BD → W7LUX | +1481 | +24.7 | 0.816 | 3-stn | Good |
| N6RFM → W7LUX | +670 | +11.2 | 0.662 | 3-stn | Good |
| AA6BD → AC0G_ND | +29 | +0.5 | 0.499 | 4-stn | Weak |
| AC0G_ND → N6RFM | +1901 | +31.7 | 0.419 | 4-stn | Weak |
| AC0G_ND → W7LUX | +1765 | +29.4 | 0.311 | 4-stn | Weak |

*Positive lag = second station lags first (wave arrives at first station earlier). AA6BD leads all other stations — consistent with northward-origin wave.*

### RESULT DIAGNOSTICS

| # | Diagnostic | 3-stn value | 3-stn | 4-stn value | 4-stn | Guideline |
|---|---|---|---|---|---|---|
| 1 | Geometry (SVR) | 7.5 | **PASS** | 1.4 | **PASS** | < ~30 |
| 2 | Residual | 13.0% | **PASS** | 26.8% | **WARN** | < ~25% |
| 3 | Pairwise corr | 0.725 | **PASS** | 0.567 | **PASS** | > 0.40 |
| 4 | Triangle closure | 39.0% | **WARN** | 63.7% | **WARN** | < ~15% |
| 5 | Phase speed | 341 m/s | **PASS** | 445 m/s | **PASS** | 100–1000 m/s |

*SVR = singular-value ratio (array conditioning). Closure flagged on both runs, suggesting one lag may not be perfectly resolved. 3-station result is preferred.*

### INTERPRETATION

Both runs agree on a NNE-origin wave (25–36°). The 3-station result (341 m/s, 1 of 5 flags) is the preferred result. AC0G_ND (North Dakota) contributes weak pairwise correlations (0.311–0.499) in the 4-station run, consistent with residual E-region contamination in the spline extraction, and elevates the triangle closure to 64%. Dropping AC0G_ND reduces flags from 3/5 to 1/5.

Triangle closure of 39% on the 3-station run remains elevated (guideline < 15%), indicating some inconsistency in the pairwise lags — likely due to the 2-hour analysis window capturing less than one full TID cycle. A tighter window centred on the clearest portion of the signal may improve closure.

The wave speed of 341 m/s is consistent with a large-scale TID. The NNE origin direction (25°) suggests a high-latitude or auroral source, consistent with typical winter LSTID generation mechanisms.

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